

Master's thesis

Master's Programme in Computer Science

Performance, Safety, and Engineering Trade-offs of Rust and C++ in Nokia RAN Control-Plane Systems

Ha Vien

September 9, 2026

Faculty of Science

University of Helsinki

Contact information

P. O. Box 68 (Pietari Kalmin katu 5)
00014 University of Helsinki,Finland

Email address: info@cs.helsinki.fi
URL: http://www.cs.helsinki.fi/

HELSINGIN YLIOPISTO – HELSINGFORS UNIVERSITET – UNIVERSITY OF HELSINKI

Tiedekunta — Fakultet — Faculty	Koulutusohjelma — Utbildningsprogram — Study programme
Faculty of Science	Master's Programme in Computer Science

Tekijä — Författare — Author

Ha Vien

Työn nimi — Arbetets titel — Title

Performance, Safety, and Engineering Trade-offs of Rust and C++ in Nokia RAN Control-Plane Systems

Ohjaajat — Handledare — Supervisors

Prof. Keijo Heljanko, Christoph Luneburg, Petteri Kotimaa

Työn laji — Arbetets art — Level	Aika — Datum — Month and year	Sivumäärä — Sidoantal — Number of pages
Master's thesis	September 9, 2026	35 pages

Tiivistelmä — Referat — Abstract

Context. Radio access network (RAN) control planes are written mostly in C++ for performance, at a well-known cost in memory safety and concurrency. Rust promises comparable performance with compile-time safety, but its fit for this latency-critical setting is unproven.

Objective. The thesis compares modern C++ and Rust on a user equipment (UE) Context Management service modelled on a real Nokia 5G base-station control-plane component (gNB-CU-CP). Average-throughput comparisons leave four dimensions open: tail latency, the C++ defect classes Rust prevents at compile time, the overhead of its safety model, and the cost of incremental adoption.

Methods. One language-agnostic specification is frozen first, then implemented twice with identical algorithms, concurrency model, allocator, and build settings. A synthetic workload is increased to overload, both versions pass the same parity tests, and the evaluation adds fault injection and a foreign function interface (FFI) prototype.

Results. (1) Safe Rust rejects all seven injected memory-safety and data-race defect classes at compile time. In C++ the same seven build cleanly and show up only at runtime, if a sanitiser catches them. (2) Seven further classes fail in both languages. On two of them C++ tooling reports what Rust does not. (3) C++ can call the Rust service through a thin C interface. All `unsafe` stays there, and any extra time on a real call is at most 10.91%. (4) A full request spends only 0.250% on language work. Passing work between threads takes the rest, so that clock cannot attribute a gap to C++ or Rust. Timed on its own, Rust's operation is 1.6× faster. (5) Below overload the full path is a few microseconds faster in Rust. The Rust versions keep up with the offered load to 448 kops/s, the C++ versions only to 392. Once neither can keep up, C++ still finishes 1.7× as much work as Rust in 20/20 mixed-order repeats with both allocators.

Conclusions. The main cost of adopting Rust is engineering effort and the learning curve, not run-time performance at nominal load. The gain is freedom from data races, not from concurrency defects in general. A nominal-load test would not expose the gap.

ACM Computing Classification System (CCS)
Software and its engineering → Software notations and tools → General programming languages
Computer systems organization → Real-time systems → Real-time system specification
Networks → Network types → Mobile networks

Avainsanat — Nyckelord — Keywords

Rust, C++, RAN control plane, tail latency, memory safety, fault injection, incremental adoption

Säilytyspaikka — Förvaringsställe — Where deposited

Helsinki University Library

Muita tietoja — övriga uppgifter — Additional information

Software study track

Contents

List of Figures iii

List of acronyms v

1 Introduction 1

1.1 Motivation . . . 2

1.2 Industrial context . . . 3

1.3 Problem statement . . . 4

1.4 Research questions and hypotheses . . . 5

2 Background: Systems Languages for Latency-Critical Software 7

2.1 RAN control-plane systems and the gNB-CU-CP . . . 7

2.2 Predictability and the two languages . . . 8

2.3 Research gap . . . 9

3 Research Methodology 11

4 Design and Implementation 13

5 Experimental Setup and Evaluation 15

6 Discussion 21

7 Conclusion and Future Work 25

Statement on the Use of AI Tools 27

Bibliography 29

List of Figures

2.1 Where the studied subsystem sits in the 5G NG-RAN 8
2.2 What the neighbouring studies control, and what they leave varying . . 10

4.1 Shared-nothing, thread-per-shard architecture 13

5.1 The two workloads, and the lifecycle the mixed one walks 16
5.2 The load sweep: tail latency and achieved throughput 17
5.3 The end-to-end operation decomposed, against what the sweep can resolve 17
5.4 Divergence past saturation 18
5.5 Where the guarantee stops: the outcome matrix 19
5.6 Resource use, and the two effects inside peak RSS 20
5.7 The C++-to-Rust FFI migration seam 20

6.1 Where Rust's compile-time guarantee stops 22

List of acronyms

Each acronym below is also written out where the text first uses it.

Acronym	Expansion
3GPP	3rd Generation Partnership Project, the body that specifies the 5G standards
ABI	Application binary interface
AI	Artificial intelligence
AI-RAN	An architecture in which AI and radio access network workloads share one accelerated infrastructure
API	Application programming interface
AWS	Amazon Web Services
CISA	Cybersecurity and Infrastructure Security Agency
CPU	Central processing unit
CWE	Common Weakness Enumeration
EU	European Union
FBI	Federal Bureau of Investigation
FFI	Foreign function interface
FIFO	First in, first out
FNV-1a	Fowler–Noll–Vo hash function, variant 1a
gNB	Next-generation NodeB, the 5G base station
gNB-CU	gNB central unit
gNB-CU-CP	gNB central unit control plane: the system class studied here
gNB-CU-UP	gNB central unit user plane
gNB-DU	gNB distributed unit
HDR	High dynamic range, of the latency histogram
IMT-2030	International Mobile Telecommunications-2030, the ITU framework for 6G
ITU	International Telecommunication Union
MPSC	Multi-producer, single-consumer
NG-RAN	Next-generation radio access network
PHY	Physical layer
RAII	Resource Acquisition Is Initialization
RAN	Radio access network
RQ	Research question
RRC	Radio Resource Control
RSS	Resident set size
UE	User equipment, an end device such as a phone or a modem
UECM	UE Context Management, the subsystem reimplemented and measured here

1 Introduction

Mobile networks are moving from specialised hardware to software on ordinary servers. Functions that once ran on fixed-purpose, vertically integrated equipment are now re-implemented as programs on general-purpose machines. The radio access network (RAN), long the most hardware-centric part of the stack, is part of that shift.

In 5G, the base station, formally the next-generation NodeB (gNB), is split into a central unit and one or more distributed units. The central unit is split again into control-plane and user-plane functions [1, 2]. Open-RAN pushes the same idea further: the RAN becomes virtualised, multi-vendor software linked by open interfaces [3]. The control plane that results, the gNB central unit control plane (gNB-CU-CP), is a large, stateful, highly concurrent program. It tracks every connected device, each of which the standards call a user equipment (UE), runs Radio Resource Control (RRC) signalling [4], and coordinates mobility and handovers, all within strict latency limits.

Load at a busy site is high. A single site can run thousands of connection-management and mobility procedures per second, each of which has to complete within the millisecond-scale budget the 5G system sets for control-plane signalling [5]. A deployed control plane may track millions of active devices at once. As the RAN moves to cloud-native, disaggregated deployments, the correctness, reliability, and timing of this software increasingly set the quality of the whole network. An outage or a tail-latency spike here is not a local defect. Every device the site serves is affected [6, 7].

This control-plane software is, today, written mostly in C++. The choice is a practical one: C++ offers the low-level control, mature tooling, and performance that latency-critical systems demand. But that performance comes with a well-known cost. C++ places the burden of memory and concurrency management on the programmer, and decades of experience show that this burden is routinely mishandled, producing use-after-free errors, double-frees, data races, and related defects that are expensive to find, dangerous in production, and responsible for a large share of security vulnerabilities in industry practice [8, 9]. Rust has emerged as the first mainstream systems language to promise an escape from this trade-off: memory safety and data-race freedom enforced at compile time on machine-checked foundations [10, 11], with performance intended

to be competitive with C++ [12]. Whether that promise holds for a telecom-grade, latency-critical control plane, where predictability under overload matters as much as raw speed, is not well understood. This thesis, carried out at Nokia, addresses the question directly: it develops two functionally equivalent implementations of a representative RAN subsystem, a UE Context Management service modelled on a real Nokia gNB-CU-CP component, in modern C++ and in Rust, and compares them under a realistic synthetic signalling workload.

Three terms recur throughout. The *gNB-CU-CP* is the control-plane software this work is about. *UE Context Management (UECM)* is the specific subsystem reimplemented and measured. *Tail latency*, the p99 and p99.9 percentiles rather than the mean, is the property that, more than average speed, decides whether such a system meets its deadlines [6, 7].

1.1 Motivation

Three trends make this comparison timely. The first is the cloud-native evolution of the RAN. Control-plane functions now run on general-purpose servers, so the software is longer-lived, more concurrent, and must meet its deadlines without the determinism of purpose-built hardware. Open-RAN disaggregation turns the base station into virtualised, multi-vendor software [3]. A latency regression or a memory-corruption defect is then a network-wide risk, and the language choice is an engineering decision.

The second trend is artificial intelligence (AI), which the network once used to tune itself and now carries as a workload in its own right [13]. International Mobile Telecommunications-2030 (IMT-2030), the 6G framework of the International Telecommunication Union (ITU), lists that second role as one of its usage scenarios [14]. The architecture forming around it is AI-RAN, in which AI and radio work share one accelerated server [15]. A control plane there no longer has the host to itself. It shares the central processing unit (CPU), memory, and cache with inference work it does not schedule. When that neighbour takes capacity, the control plane saturates sooner: it stops keeping up with demand at a lower load than it would alone. Chapter 5 finds the widest gap between the two languages past that point.

The third is the shift toward memory-safe languages. Large software organisations report that the majority of their serious security vulnerabilities stem from memory-safety errors, and that adopting memory-safe languages reduces them [8, 9, 16, 17].

Independent work argues that memory safety now needs standardised requirements if it is to be procured at all [18]. Cybersecurity agencies urge a move away from C and C++, and the European Union (EU) Cyber Resilience Act adds legal duties from September 2026 and December 2027 [19, 20, 21]. The Act names no programming language. Direct evidence for a RAN control plane is still missing. Providing it is the motivation for this work.

1.2 Industrial context

This thesis is carried out at Nokia. The company is one of the small number of vendors that build, maintain, and evolve production RAN software end to end. The five largest suppliers took 96% of worldwide RAN revenue over the first three quarters of 2025, with concentration by Herfindahl–Hirschman Index at a ten-year high [22, 23]. The subsystem it studies, UECM, is modelled on a real gNB-CU-CP component rather than a textbook example.

In the architecture of the 3rd Generation Partnership Project (3GPP), the gNB-CU-CP is the control-plane half of the central unit. It hosts RRC and manages the UE context lifecycle [1, 4], under the millisecond-scale budgets the 5G system sets for control signalling [5]. Chapter 2 places it in the wider architecture. Because it is written in C++, it inherits that language's performance and its memory-safety and concurrency hazards. The three trends of Section 1.1 meet here. Software of this class runs on live operator networks. The language choice is a production concern.

UECM is small enough to reimplement faithfully and reason about in full. It still carries the properties that make control-plane code hard. The implementation follows a public, frozen specification, not proprietary code, so readers outside Nokia can reproduce the comparison.

A decision with a generational horizon. A language choice in software of this class outlives the generation it was made for. 3GPP has allocated Release 20 to the 6G studies and Release 21 to the normative work. The ITU asks for technology proposals in early 2029 and submitted specifications by mid-2030 [24]. That work covers migration from 5G and interworking with it. A control plane written now is maintained alongside the next generation, not retired. IMT-2030 still names security and resilience as a design principle and still carries the ultra-reliable low-latency scenario in strengthened

form [14].

The measurements here use a current-generation subsystem and published requirements. No claim is made about requirements that have not been written.

A concrete failure mode. Consider a mobility storm in which many UEs hand over at once. A UE context freed on release while an in-flight signalling message still references it is a use-after-free. In C++ that compiles and may pass ordinary tests. It can then corrupt another UE's state or crash the worker under high concurrency. The same defect can show up as a tail-latency spike instead of a crash. In safe Rust the design does not compile: ownership and borrowing reject the dangling reference. That is why this thesis measures both *safety* (which defects are prevented, and when) and *predictability* (tail latency under overload), not average throughput alone.

1.3 Problem statement

Two properties of this domain make the language choice consequential. First, manual memory and concurrency management in C++ exposes the software to defects such as use-after-free, double-free, data races, and iterator invalidation, which are silent at compile time and costly to find and fix at telecom scale. Second, what governs whether RAN deadlines are met is not average throughput but worst-case and tail behaviour under load and overload [6], a property that is easy to regress and hard to measure correctly. Rust promises to remove the first class of defect at compile time while claiming performance competitive on the second, but that claim rests mostly on general-purpose benchmarks rather than on stateful, concurrent, latency-critical telecom workloads.

Bringing these threads together, the concrete problem this thesis addresses is the following. Telecom vendors are under growing pressure, both technical and increasingly regulatory, to adopt memory-safe languages, yet for the RAN control plane they lack *direct, controlled* evidence on which to base that decision. It is not established whether a Rust implementation of a stateful, concurrent control-plane subsystem can match modern C++ in *tail latency and predictability under overload* (as opposed to average throughput), whether Rust's compile-time guarantees actually eliminate the specific defect classes that affect the existing C++ code, and what incremental adoption *costs* inside a large C++ codebase. The available evidence is largely general-purpose and

average-case [12, 25], leaving these three questions (predictability, prevented-defect scope, and migration cost) open for a representative subsystem. Answering them under controlled, reproducible conditions is the problem this thesis sets out to solve.

1.4 Research questions and hypotheses

The thesis is organised around four research questions, ordered to put predictability and migration ahead of raw average performance:

RQ1 How do Rust and modern C++ compare in latency predictability (median, p99, p99.9 and p99.99 tail latency, and the maximum) for a RAN control-plane workload under increasing load and overload?

RQ2 Which classes of memory-safety and concurrency defects that occur in C++ are eliminated at compile time in the Rust implementation, and at what stage and cost are the remaining ones detected?

RQ3 What is the overhead, if any, of Rust's safety model (bounds checks, ownership-driven copies, abstraction cost) on throughput, CPU utilisation, and memory behaviour?

RQ4 What are the practical implications (effort, foreign function interface (FFI) interoperability, and risk) of adopting Rust incrementally within an existing C++ RAN software ecosystem?

Three prior expectations are committed before any measurement, so the evaluation can confirm or refute them. **H1:** For this workload the two languages are performance-equivalent in the operationally relevant region below saturation, with any difference confined to the far tail. **H2:** Safe Rust prevents at compile time the memory-safety and concurrency defect classes that C++ leaves to best-effort, after-the-fact detection. **H3:** The run-time cost of C++-to-Rust interoperability is negligible relative to the cost of the operations invoked, so that the true cost of incremental adoption is engineering effort rather than performance. The defect catalogue behind RQ2 spans classes safe Rust is expected to prevent and classes it is expected not to, so the study can locate the boundary of the guarantee instead of only confirming it. Chapter 6 revisits all three against the evidence.

The boundary RQ2 sets out to locate has been estimated once already at the scale of a whole weakness taxonomy. Every base entry of the Common Weakness Enumeration (CWE) has been mapped onto what safe Rust would prevent, and weighting those classes by the vulnerability history of 68 open-source C projects puts 58.2% of the classified vulnerabilities into classes safe Rust would have made virtually impossible [26]. The estimate is a best case by construction. It assumes the converted code stays safe, and it reads prevention from the weakness class rather than from a compiler that refused to build a particular program. RQ2 asks the same question of programs that are compiled and run, which is what lets it report at what stage and cost the classes outside the guarantee are caught.

2 Background: Systems Languages for Latency-Critical Software

2.1 RAN control-plane systems and the gNB-CU-CP

The next-generation radio access network (NG-RAN) architecture splits the gNB into a central unit (gNB-CU) and distributed units, with the control plane (gNB-CU-CP) handling RRC and connection management for each UE [1, 2, 4]. Such control-plane systems must satisfy demanding non-functional requirements: very high event rates (UE mobility, signalling storms), strict and predictable latency bounds that the 5G service and RAN specifications quantify as key performance indicators (KPIs) [5, 27], long-lived software with continuously evolving complexity, and heavily concurrent workloads. In these systems the worst-case and tail behaviour, rather than the average, determines whether deadlines are met, which motivates the predictability focus of this thesis.

Functionally, the gNB is disaggregated along two independent axes. The first is a *central/distributed* split: a central unit (gNB-CU) hosts the higher layers of the protocol stack, while one or more distributed units (gNB-DU) host the lower, more time-critical layers close to the radio. The second is a *control/user-plane* split inside the central unit, separating the control-plane function (gNB-CU-CP) from the user-plane function (gNB-CU-UP). These parts communicate over standardised interfaces: NG towards the 5G core, F1 between the CU and the DU, and E1 between the CU-CP and the CU-UP [1, 2, 28]. Open-RAN initiatives standardise these interfaces further so that each part can be supplied by a different vendor and run as software on commodity hardware [3]. Figure 2.1 places the subsystem studied in this thesis, UE Context Management (UECM), inside the gNB-CU-CP: it owns the per-UE control-plane state and mediates the RRC signalling and mobility procedures, which is why its correctness and timing behaviour are representative of the control plane as a whole.

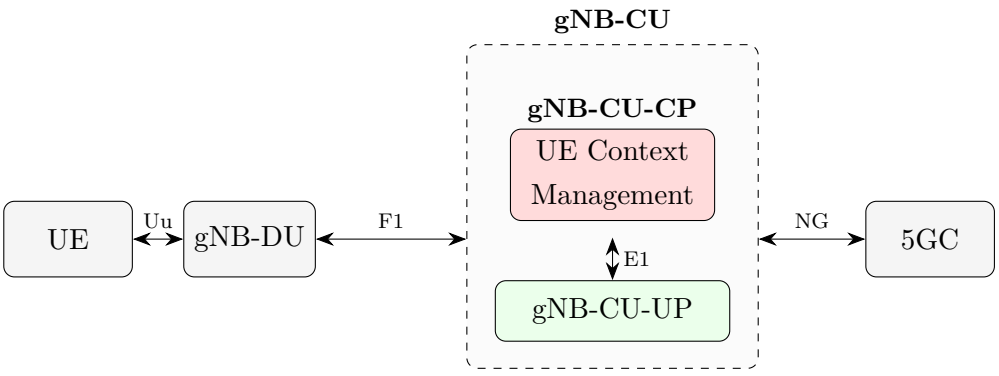


Figure 2.1: Where the studied subsystem sits in the 5G NG-RAN. The gNB is split into a distributed unit (gNB-DU) and a central unit (gNB-CU), and the central unit is split again into control-plane (gNB-CU-CP) and user-plane (gNB-CU-UP) functions, connected by the standardised F1, E1, and NG interfaces. UE Context Management (highlighted) is the per-UE control-plane state manager inside the gNB-CU-CP that this thesis reimplements and measures.

2.2 Predictability and the two languages

Four strands of background bear on the comparison, and this section states each as the position the study inherits from it. Predictability, not average throughput, is the governing property of a latency-critical service, so the quantity to measure is the tail of the distribution and not its mean, and measuring it correctly constrains the harness as much as the metric: an open-loop schedule is required, because a closed-loop one stops issuing requests while a stall is in progress and so never records the latency of the stall.

C++ supplies the control the domain demands and leaves memory and concurrency safety to the programmer, with sanitisers and review standing in for a guarantee. The standards committee's work on enforced safety profiles is neither standardised nor generally implemented. Rust's ownership and borrowing discipline rules out lifetime and aliasing errors at compile time, and its `Send` and `Sync` traits rule out data races. That is a narrower claim than the phrase "thread safety" suggests. It says nothing about lock-acquisition order, about the atomicity of a composed operation, or about leaks, and it is suspended wherever a program writes `unsafe`. Both languages can express the same concurrency structures, and a comparison that lets the structure vary with the language measures the structure. The two meet through a C application binary interface, which is the mechanism any incremental migration has to use and the reason a seam concentrates whatever `unsafe` a migration needs.

2.3 Research gap

The preceding sections set up the gap. The RAN control plane is a soft-to-firm real-time, highly concurrent domain where tail behaviour under overload, not average throughput, governs the quality the network delivers. C++ supplies the control and performance the domain demands, but leaves whole classes of memory and concurrency defects to best-effort detection after the fact. Rust promises to prevent those classes at compile time at no run-time cost, and offers a credible incremental adoption path through the C application binary interface (ABI).

The domain is not unstudied, and the volume of prior work is easy to underestimate because almost none of it is indexed. At least five vendor-hosted master's theses, all carried out at Ericsson, have compared candidate languages for RAN and telecom software. One weighs runtime cost against development efficiency across C++, Rust, and Go [29]. One asks whether Rust could replace C in RAN node management [30]. One builds a thread-per-core packet processor on asynchronous Rust and measures it against the Data Plane Development Kit (DPDK) application it mirrors [31]. One measures median and 99th-percentile latency for control-plane procedures across five asynchronous programming frameworks [32]. The last builds matched C++ and Rust implementations of an Abstract Syntax Notation One (ASN.1) decompiler and compares them statistically [33]. All five are directly relevant, and two report differences unfavourable to Rust.

Three of them vary the language together with the runtime and the concurrency model, so a measured difference cannot be attributed to the language rather than to the framework built on it [29, 31, 32]. The effect sizes reach a factor of five, which is an indication of how much of the measurement the runtime owns. The remaining two hold the design closer to fixed but report execution time and peak memory rather than tail behaviour under rising load [30, 33], and the closest of them, Emberg and Wadsten, measures a stateless transformation rather than a stateful, concurrent subsystem. Figure 2.2 sets the neighbouring studies against the properties a controlled answer would require, and the pattern it shows is that each holds some of them and none holds all.

The concentration of that evidence is itself part of the gap. All five studies come from one vendor, and their conclusions have never been tested against another organisation's software, targets or conventions, so what is known about this question is what one company's frameworks report about it. The published record on the other side is

emptier: there is no study of Rust among the indexed publications of the vendor hosting this work, and the nearest peer-reviewed position from that side argues for making memory-error sanitisation practical rather than for replacing it with a compile-time guarantee [34], which is the strategy this thesis tests against. The paragraphs below compare the five with this study directly.

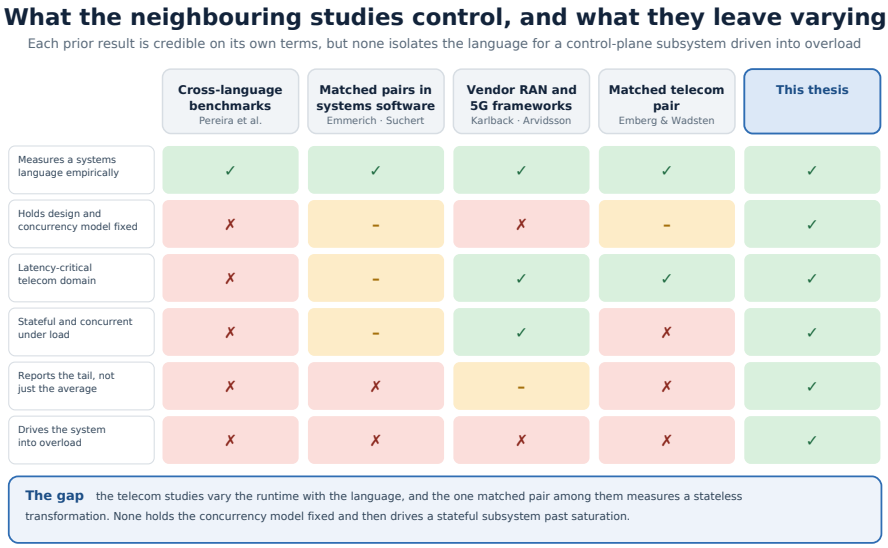


Figure 2.2: What the neighbouring studies control, and what they leave varying. Each row is a property that a controlled answer to the research questions would need. Each column is a study. The general cross-language comparisons measure the right kind of object but the wrong regime, reporting averages on general-purpose workloads. Matched pairs in systems software hold the design fixed, but outside this domain and below saturation. The vendor RAN and 5G framework studies reach the right domain, and one of them reports the tail, but in all of them the language varies together with the runtime and the concurrency model. The one matched telecom pair, Emberg and Wadsten, controls the design but measures a stateless transformation. Holding the design and the concurrency model fixed for a stateful, concurrent subsystem and then driving it past saturation is what this thesis adds. The contribution is the control, not a new measurement technique.

What is therefore missing from the literature is *direct, controlled* evidence for a representative control-plane subsystem: Whether Rust actually holds its tail latency and predictability under overload against modern C++ *when the concurrency model, algorithms, and data structures are held fixed*, exactly which defect classes it eliminates and at what stage the remainder are caught, what its safety model costs in throughput and resources, and what incremental adoption costs in practice. Closing this gap is the purpose of the four research questions. The next chapter grounds them in the specific subsystem, defect classes, and timing properties at stake, and sets out the methodology by which they are pursued.

3 Research Methodology

The approach is empirical and constructive. Instead of comparing two systems that already exist and differ in many ways, the study builds a matched pair of implementations from one frozen specification. The language is then the only systematic difference between them. Both use an identical thread-per-shard concurrency model. That structuring itself changes tail latency [35].

RQ1 and RQ3 are answered with an open-loop load sweep from light load into overload on pinned cores. Open-loop means the generator keeps to its send schedule instead of waiting for replies. Latency is recorded in identically configured high-dynamic-range (HDR) histograms [36]. Resource use is sampled through the operating system. The reported sweep runs on a dedicated AMD EPYC 7443, simultaneous multithreading off, frequency governor fixed at performance, with `nohz_full` on the pinned cores and no hypervisor. That host resolves the overload divergence, the resource comparison, and the far tail below saturation. The same design was first measured on a Intel(R) Xeon(R) CPU E5-2680 v4 guest (kvm) that could not resolve the far tail: there the widest interval on one variant's p99 exceeded the spread between all four by a factor of five, so no ordering could be read however many repeats were run. A third sweep on a second bare-metal AMD EPYC 7443P uses a different shard count, load grid, repeat count and toolchain, so it tests whether the findings survive a change of design rather than a change of machine.

RQ2 is answered by controlled fault injection [37]. A fixed catalogue of defects is written in both languages. The expected outcome of each one is recorded in advance. A surprise then shows up as a discrepancy, not as something explained away afterwards.

RQ4 is answered by a working prototype in which C++ drives the Rust core through a foreign-function interface, a thin seam of code where the two languages meet. Cross-language memory-management defects concentrate at that seam [38]. It yields three measurements: How large the seam is, how much `unsafe` code it needs, and what one call across it costs.

Fairness is the central experimental control. The specification is frozen. Allocator and optimisation settings are matched. Every point where the two languages had to differ is written down in a forced-idiom log. Reporting follows the guidance that a single figure

invites conclusions run-to-run variation does not support [39]. Medians come with nonparametric bootstrap confidence intervals [40]. Exact paired permutation tests are corrected for multiple comparisons by the Holm procedure [41]. All data is synthetic, so no data-protection concerns arise.

4 Design and Implementation

Nothing is implemented until the specification is frozen. It fixes the public API, the UE state machine, the data structures and their algorithmic strategy, the locking and sharding granularity, and the concurrency model. Both implementations conform to it. The comparison then measures the language, not a difference of design or of skill. The specification is itself a deliverable. It is the study's main control against implementation bias.

One concurrency model is used on both sides, shared-nothing and thread-per-shard. That structuring is known to change tail latency by itself [35]. The identifier space is partitioned into shards. Each shard is owned by a single worker thread that alone reads or mutates it, so no lock guards context state. Callers submit a request to the owning shard's bounded queue and wait on a single-use completion handle. The queue blocks when it is full. That is deliberate. A drop policy would hide overload, and this study needs overload to appear as latency. The only shared mutable state either version permits is one atomic counter. No control flow depends on its value. Figure 4.1 shows the resulting data flow, with one multi-producer, single-consumer (MPSC) queue per shard that its worker drains first in, first out (FIFO).

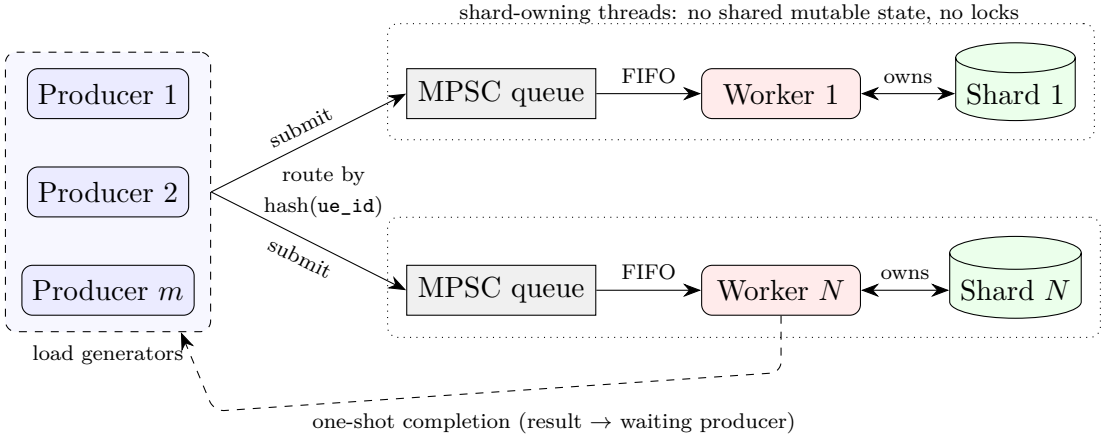


Figure 4.1: Shared-nothing, thread-per-shard architecture. Producers submit request messages to a shard's bounded, blocking queue, selected by a pinned hash of the `ue_id`, and wait on a single-use completion handle. Each shard is read and written by exactly one worker thread, so there are no locks on context state, and blocking on a full queue is what makes overload observable as tail latency.

Fairness is enforced below the design as well as in it. Both builds link the same

`jemalloc` and keep a system-allocator build as a control. Both compile at matched optimisation. Both record latency through one HDR histogram algorithm configured identically [36]. Hashing is pinned on both sides, SplitMix64 for shard assignment and FNV-1a, the Fowler–Noll–Vo hash, inside the map. C++'s default hash of an integer is typically the identity function. Rust's is cryptographically strong. Left as they came, the two standard libraries would have been the measurement.

Where the languages cannot satisfy a requirement the same way, the divergence is written into a forced-idiom log and then measured. Rust takes its bounded blocking queue from `sync_channel`. C++ through C++23 has no such primitive. It assembles one from a mutex and two condition variables. C++ has no payload-carrying enumeration in this idiom either. A request becomes a tagged struct holding every argument field any operation might need, 160 bytes against Rust's 96. The caller answers through a `variant` it has to unwrap. The net result is a Rust core 18% longer in code lines whose total cyclomatic complexity is 15% lower [42]. The branchiest function in either core is the C++ worker loop that dispatches that `variant`. The Rust library contains no `unsafe` at all.

Parity is tested and not assumed. Language-neutral vectors run in both implementations and their transcripts are diffed operation by operation. A cross-diff cannot tell "both correct" from "both wrong in the same way". A third implementation in Python acts as an independent oracle. It has to reproduce all 223 hand-derived expectations before it may generate any of its own. A campaign of roughly 793 000 generated operations found no divergence. The transition matrix is enumerated exhaustively instead of sampled. The C++ side runs under AddressSanitizer, ThreadSanitizer and UndefinedBehaviorSanitizer. The Rust side runs under Miri. Line coverage is reported. A clean sanitiser result is therefore bounded by the code that actually ran.

5 Experimental Setup and Evaluation

The workload is synthetic and spec-justified. A deterministic driver walks each UE through the connection lifecycle the 3GPP specifications define. Its draw order is pinned, so the per-operation-kind counts come out identical in both languages for a given seed. That doubles as a parity signal, and Figure 5.1 shows the two workloads and the procedure sequence the mixed one walks. Each sweep point is repeated 20 times. The four language and allocator variants are interleaved inside every repeat, so thermal or background drift falls on all of them alike. Every point is a median with a bootstrap 95% confidence interval [39, 40]. The representative operating point is fixed at 56 kops/s and not derived. The rule this study began with, the busiest swept load holding p99 under 5 ms, located the virtualised host's noise floor and on a quiet host climbs until it collapses onto the saturation knee, the load beyond which throughput stops following demand, so it is kept as a diagnostic only. The measured throughput plateau is 392 kops/s, the lowest of the four, reached at an offered 392 kops/s. All of it runs on a dedicated AMD EPYC 7443 pinned to cores 2–17. The frequency governor is fixed at performance with boost disabled. The kernel runs tickless on those cores. There is no hypervisor. Isolating the cores with `isolcpus` is omitted on purpose, because it would collapse the pinned set onto one core. The same design was first measured on a Intel(R) Xeon(R) CPU E5-2680 v4 guest that could not resolve the far tail. A second bare-metal host at a different shard count confirms that the directions survive a change of design.

Below saturation the four variants sit within 2 μs of one another at the median, 2.1%. The paired test resolves a difference of 1.3 μs in Rust's favour under both allocators. Figure 5.2 draws the sweep behind that, tail latency and achieved throughput against offered load. Higher up the distribution the ordering can now be read. The widest interval on a single variant's own p99 spans 1 μs against 2 μs of spread between all four, so the uncertainty about one is smaller than the gap between them. On the virtualised host the same two quantities were 1480 μs against 293 μs, the wrong way round. The far tail resolves here too, from the repeats' histograms pooled into 3 360 051 to 3 360 084 operations per variant, and both Rust variants sit below both of the others by up to

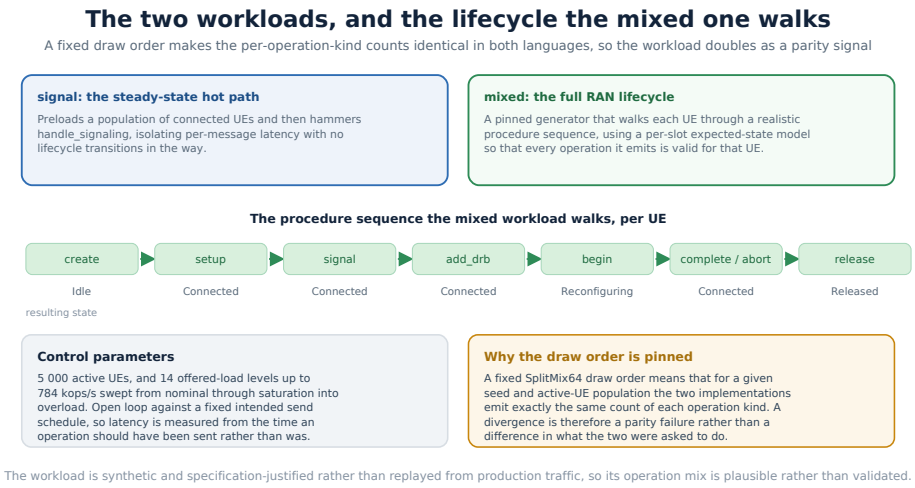


Figure 5.1: The two workloads. The *signal* workload isolates the steady-state hot path, while the *mixed* workload walks each UE through the full RAN procedure sequence, with a per-slot expected-state model ensuring that every operation it emits is legal for the state its context is actually in. Because the SplitMix64 draw order is pinned, a given seed and active-UE population produce identical per-operation-kind counts in both languages, so a divergence is a parity failure rather than a difference in what the two implementations were asked to do.

87 μs. A second bare-metal host at another shard count agrees, running 147–228 μs for the C++ variants against 88–115 μs for the Rust ones, without overlap. A second limit is structural. No host would fix it. An end-to-end operation is 0.250% the work the language performs. The rest is a cross-thread handoff charged to the operating system [43]. This sweep cannot bound a language effect in either direction, which Figure 5.3 sets against what it can resolve. Measured on its own the bare operation is 1.6× faster in Rust. The likely cause is the standard hash map's layout, not anything about safety.

Once the offered rate outruns the service, two differences appear. They stop meeting the offered rate at different loads: the Rust arms hold to 448 kops/s and the C++ arms to 392, so at 448 kops/s Rust still serves 448 kops/s where C++ has collapsed to 162. That load level is described and not tested, because the operating point was found after the data was in hand. Deeper, at 784 kops/s offered, the ordering reverses: C++ sustains 147 kops/s where Rust sustains 84, a factor of 1.7, in 20/20 repeats under both allocators with the Holm-corrected p at the floor of the whole test family. The slower variant is not simply doing less work. It draws 1.8 times the CPU per completed operation, which is the signature of effort spent without completing anything. Isolating the bounded queue and the completion handle on their own does not reproduce that

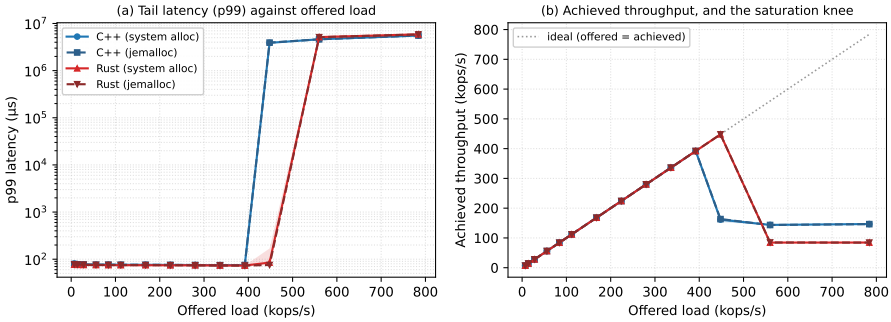


Figure 5.2: The load sweep, both panels against offered load. **(a)** Tail latency (p99) for the four build variants on a log axis, flat across the whole tracking region and then rising by four orders of magnitude in one step of the ladder. **(b)** Achieved throughput against the dotted ideal, which the C++ curves leave after an offered 392 kops/s and the Rust curves after 448, so 448 kops/s is the only load in the sweep at which the two languages differ in *throughput* with Rust ahead. Lines are medians over 20 repeats and shaded bands are bootstrap 95% confidence intervals in both panels.

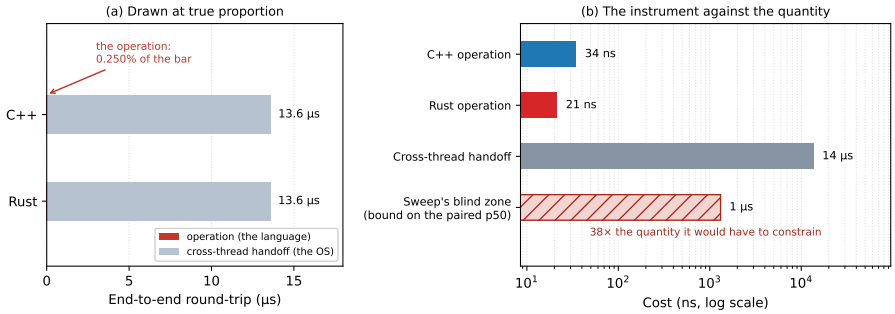


Figure 5.3: Where an end-to-end operation's cost goes, and why the sweep cannot bound a language effect. **(a)** The round-trip at true proportion, in which the operation the language performs is 0.250% of the bar and the remainder is the cross-thread handoff. **(b)** The same quantities on a log axis against the smallest paired difference the sweep can resolve, shown as a blind zone that both operation measurements fall inside by a factor of 38.

ordering at significance, so the mechanism sits in the wider handoff path. This is the largest language effect in the tested family. A comparison that stopped at nominal load would have concluded the opposite. Figure 5.4 sets what each variant still completes beside what it spends completing it.

Safety is measured by injecting 14 defects. Seven come from inside the region Rust claims to govern. Seven were chosen because safe Rust is expected *not* to prevent them. Every one of the first seven is a compile error in Rust. It cannot be written at all. In C++ each is valid code. The defect appears only if a test drives it, and no single sanitiser covers them. The use-after-move is the case where no sanitiser reports

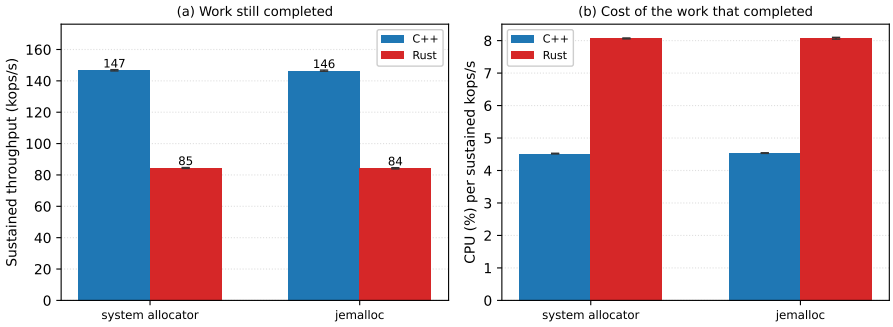


Figure 5.4: Behaviour at 784 kops/s offered, the deepest swept load at which every variant has stopped meeting the offered rate. Bars are medians over 20 repeats with bootstrap 95% intervals, grouped so that the two bars inside a group differ only in language. **(a)** What each variant still completes, where C++ sustains a factor of 1.7 more than Rust. **(b)** Total CPU divided by the work that actually completed, which is what rules out the reading that the variant delivering less is simply doing less. It spends 1.8 times as much CPU per sustained operation, and the CPU figures are summed across threads.

anything: a moved-from object is valid but unspecified, so there is nothing to detect and the identity is simply lost. All seven boundary cases compile in safe Rust and misbehave. On two of them the C++ tooling is the one that reports. LeakSanitizer flags the reference cycle that Rust leaks in silence. UndefinedBehaviorSanitizer flags the overflow that Rust defines to wrap. ThreadSanitizer catches the lock-order inversion on some runs and not others, at a rate that is itself unstable. It builds its lock-order graph from acquisitions that succeed. A run whose first inversion blocks never closes the cycle. Two Rust model checkers do report both original boundary defects, Loom exploring interleavings soundly and Shuttle sampling the ones Loom cannot reach [44], but only after each was re-imported against the checker's own primitives and shrunk to model size. Miri reports the seam double release, the one class that safe Rust rejects and unsafe Rust does not, but it interprets Rust and cannot execute the C++ host the defect ships behind. That half of the boundary is caught at test time, and for the concurrency cases only after a rewrite. Figure 5.5 draws the whole catalogue as an outcome matrix.

Resource use separates the variants where latency could not. Two effects of comparable size sit in the resident set. The Rust build costs 1.4–2.5 MiB more peak resident set than the C++ build under either allocator. Linking `jemalloc` at all costs 1.5–2.4 MiB in either language. Both are real, and Figure 5.6 puts the pair beside the peak resident set size (RSS) they decompose. The language effect keeps its direction and order of

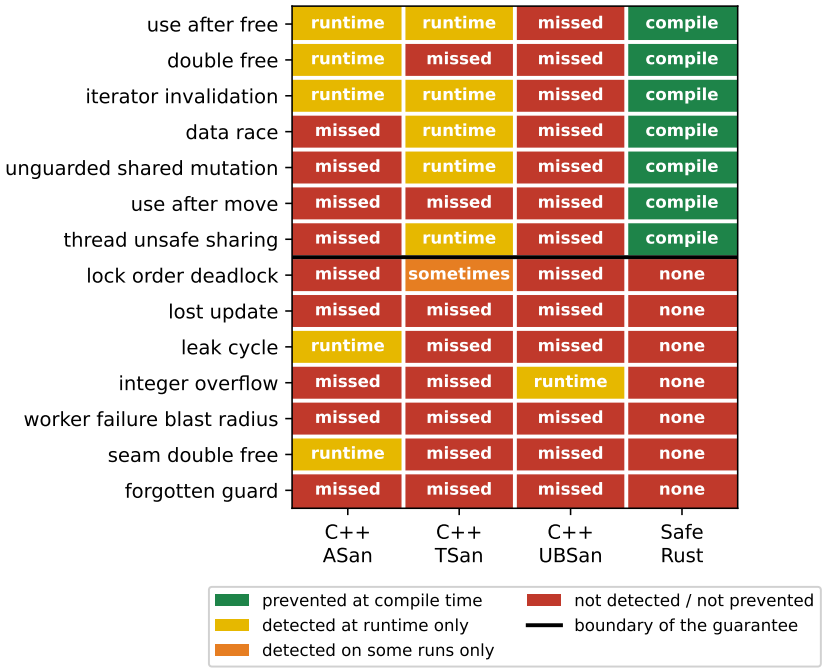


Figure 5.5: The fault-injection outcomes drawn as a matrix, generated from the taxonomy log the runner writes. Above the rule the Rust column is a single colour, because every defect there is a compile error and none can be written, while the C++ columns are a patchwork that no one sanitiser covers and the use-after-move row is covered by none of them. Below the rule the Rust column joins them, on the reference-cycle leak and the release-build overflow the C++ tooling is ahead, and the lock-order deadlock is the one cell where a tool reports intermittently, ThreadSanitizer flagging the inversion in 5 of 12 runs while the program deadlocks on every one.

magnitude on all three machines measured. The allocator's was an order of magnitude larger on the virtualised host this study used first, so their ratio is a property of the machine. The language difference is a fixed cost, not a per-context one, and the quiet host locates it. `jemalloc`'s own accounting there puts +2.07 MiB more in arena bookkeeping and −0.16 MiB *less* in live allocation, and its resident bytes are −1.92 MiB lower while the process peak is +1.74 MiB higher. So the footprint sits outside the allocator, in the runtime a Rust process carries before it allocates. Counters at the operation scope explain the timing result instead: the Rust build retires 11% fewer instructions in 15% fewer cycles. CPU inverts with the regime. It is marginally lower for Rust while the service copes and much higher per completed operation once it does not. The migration seam is small on every axis measured: 11 C entry points in 231 lines of Rust. All raw-pointer work is confined to 9 functions. The core is left free of `unsafe`. No cost is detected against the same operation called natively. That bounds any per-operation tax at 10.91%, and Figure 5.7 shows the structure that bound belongs to.

Three adoption costs beyond the seam were measured as well. Two of them contradict the received objection. 54% of the Rust clean build is third-party code. The project's own source compiles in 0.8 times the CPU `g++` needs. The 6.8× binary-size gap is a linkage policy that reverses when the policy is held fixed. The compared component resolves 0 crates and contains 0 `unsafe` sites.

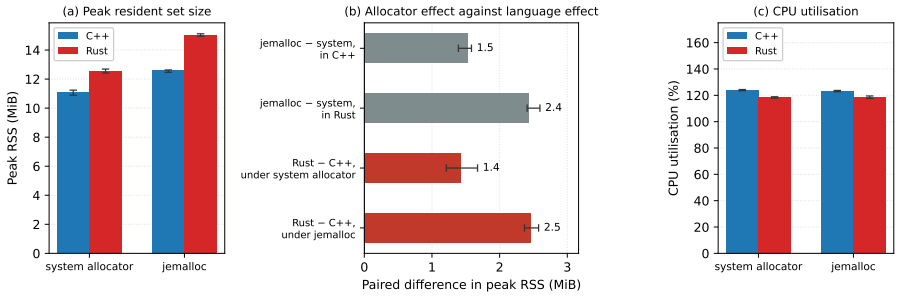


Figure 5.6: Resource use at a sustained offered load of 56 kops/s, grouped by allocator so that the two bars in a group differ only in language and the two groups differ only in allocator. Medians over 20 repeats with bootstrap 95% intervals. **(a)** Peak resident set size. **(b)** The two effects it contains, each as a paired difference over the same repeats, which on a shared axis are of comparable size here and were not on the host this study used first. **(c)** CPU utilisation, summed across threads, where the difference runs the other way and is small.

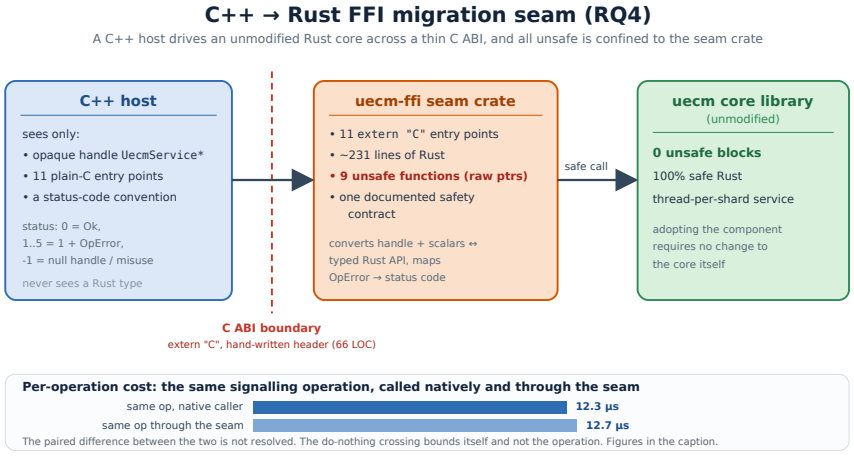


Figure 5.7: The C++-to-Rust FFI migration seam. The C++ host sees only an opaque handle, 11 plain-C entry points, and a status-code convention, while a thin `extern "C"` seam crate concentrates all of the raw-pointer `unsafe` in 9 functions and calls the unmodified Rust core, which remains free of `unsafe`. The band along the bottom is the comparison the argument rests on, the same signalling operation called natively at 12.3 µs and through the seam at 12.7 µs, whose paired difference is not resolved and is bounded at 10.91% of one operation. The do-nothing crossing costs 2.54 ns and bounds only itself.

6 Discussion

What a deployment should do with RQ1 depends on where it expects to sit, not on a single ranking. A service sized to run below every arm's knee should prefer the arm that holds its offered rate further. One that a signalling storm will drive past every knee should prefer the arm that still completes more work afterwards. Those two preferences point at opposite languages, which is why Chapter 5 reports both halves rather than a winner. The closest prior work measures this quantity for 5G control-plane procedures. It finds the same shape and traces it to stackless `async/await` [32]. That attribution is accepted here. The shape appeared with no asynchronous runtime on either side and the concurrency model held fixed. A runtime cannot be the only place such an effect originates. The prediction registered in advance also named where to look first, bounds and overflow checking. That part of it was wrong. Recording the miss is worth more to the next study than quietly substituting the explanation that survived.

On safety the difference is one of kind. The boundary is reported as a first-class result, not as a caveat. A compiler that refuses to build a program rules the defect out of every execution. A sanitiser rules it out only of the executions an instrumented test happens to take. What is absolute there is the execution coverage, not the breadth of the catalogue, and Figure 6.1 names the mechanism on each side of that line. Rust reasons about aliasing and not about acquisition order. A `Mutex` guard whose lifetime the borrow checker tracks perfectly says nothing about whether two threads take their locks in a compatible order [45]. The lost update escapes ThreadSanitizer for the same reason it escapes the borrow checker. Where every access is correctly synchronised, there is no unsynchronised access to find. Adopting Rust retires one category of concurrency defect and leaves untouched the category that fails silently. A migration plan reading “memory-safe” as “concurrency-safe” would mis-budget its review effort. One case reaches past the injected programs and narrows the credit due on both sides. A specification gap let an operation be submitted outside the service's running window. It survived every test, every parity vector and all three sanitisers because nothing ever called that path. That is the weakness of detection conditional on coverage. What turned undefined behaviour into a diagnosable failure on the Rust side was an ordinary bounds check, not a proof. The property worth budgeting for is “checked by default”, not “ruled out at compile time”. The root cause belongs to the method. A

prohibition stated without a mechanism behind it leaves each implementation to fail in its own idiom.

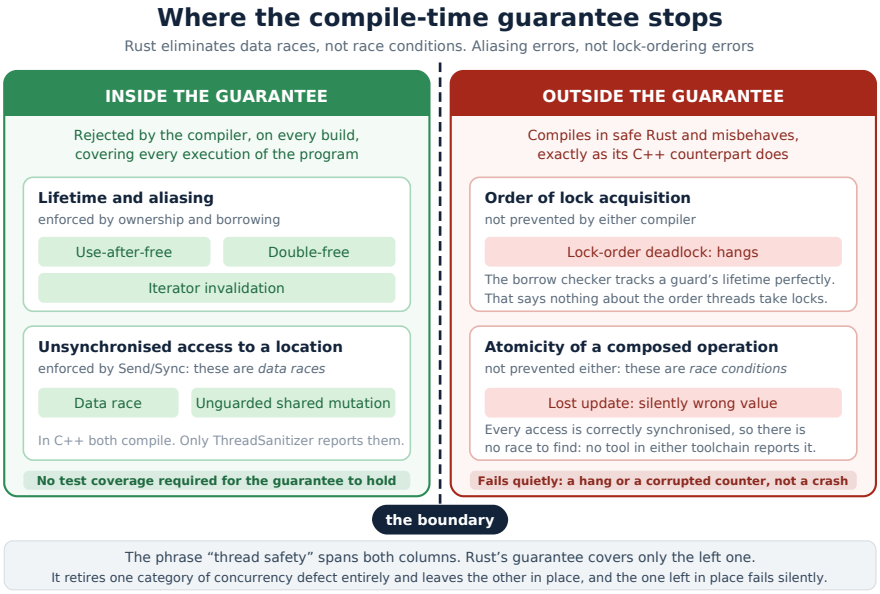


Figure 6.1: Where the compile-time guarantee stops. On the left are the classes safe Rust rejects outright, grouped by the *mechanism* that rejects them rather than by defect family, which is ownership and borrowing for lifetime and aliasing errors and the `Send`/`Sync` discipline for unsynchronised access to a location. On the right are the classes the catalogue included to find the edge, which neither compiler prevents. The diagram names one class per mechanism rather than the whole catalogue, the phrase "thread safety" spans both sides of the line, and the line itself moves once more than the diagram can show, because a program that writes `unsafe` crosses it voluntarily.

RQ3 splits by resource and the splits do not point the same way. RQ4 comes out cheap in the machine and expensive in the people. The memory penalty reproduces under both allocators and on a second machine with a different compiler. It is credible beyond its p-value. Peak resident set is a run-level maximum insensitive to the jitter that destroys a p99. It is a few MiB of fixed cost, and sized against the allocator's own effect the two are of the same order on both bare-metal hosts, interleaved on the reported one and with the allocator ahead on the other. On the virtualised host the allocator's was an order of magnitude larger. The language effect travels to other machines and the ratio does not. The quiet host also locates the cost. `jemalloc`'s own accounting puts it *outside* the allocator, because the Rust build holds less live data and fewer resident bytes there and still peaks +1.74 MiB higher. What a Rust process carries before it allocates is the standard library and the runtime. Which of its pages hold the difference is not attributed to a call site. One close study reports the opposite sign, with Rust lowest and the gap widening as concurrency rises [29].

The two are compatible. Their C++ runs on Boost.Fibers, which allocates a stack per fiber. Their Rust runs on Tokio, where a task is a small heap-allocated future. What they measured is the per-context cost of two concurrency runtimes. Two further studies agree with the direction found here [30, 46]. On migration the dominant cost is human [47, 48]. Confining `unsafe` to the seam bounds what has to be reviewed, not what can be corrupted. Host and component share one address space [49, 50]. The build and footprint axes point the same way as the seam. An organisation worried about either should audit its dependency budget before blaming the compiler. The one axis where C++ leads outright is the edit-compile loop for changes that stay inside a translation unit.

The three hypotheses were committed before any measurement. Stating their disposition is what separates a test from a description. H1's median clause holds at the representative load. Its far-tail clause does not, and overload, which the wording never named, is where the languages actually part. The reported campaign now resolves that tail, in Rust's favour, and a second bare-metal host at another shard count agrees on the direction. H1 was right about where a below-saturation difference would show up and wrong that it was the only place. It was also not well posed. "Performance-equivalent" reads as one quantity and names at least three. H2 holds without exception for the seven injected classes of its kind. It fails in the looser reading its wording invites. It is accepted for memory safety and data-race freedom, rejected for concurrency correctness at large, and it is a claim about *safe* Rust only: the seam case shows the same class returning inside `unsafe`. H3 is supported by the bound on a cost the measurement could have missed, not by the failure to detect one. Three threats stay load-bearing behind all of it. One author wrote both implementations. The parity suite bounds behaviour and not performance. The end-to-end instrument measures the design faithfully and the language barely at all. Single-author bias is documented and not hypothetical: the closest matched pair in this industry reversed its own verdict across two benchmarking sessions, comparable after a code review and then Rust worse by 85–91% after further optimisation [33]. The overload divergence is replicated here with its mechanism narrowed and not closed, and no peer-reviewed study reports the same reversal.

The guidelines are the practical form of all this. Each carries the limit of the finding behind it. Choose a first component for the narrowness and the scalarity of its interface, not for its size. The redesign of a rich interface is the schedule risk. The boundary itself is not. Keep the C ABI in its own crate and the core free of `unsafe`. That keeps

unsafety out of the dependency chains it otherwise propagates through [51], and makes the reviewable surface a countable thing. Leave the concurrency review budget where it is. Prefer a shared-nothing structure in either language. The strongest concurrency result here belongs to the design and to neither compiler. Plan the learning curve as a training cost and instrument it. The second component can then be scheduled on measured effort. Measure the component as well as the round-trip. Sweep past the point where the service stops coping. That is the guideline this study would most like to have followed from the start.

7 Conclusion and Future Work

The thesis compared modern C++ and Rust on a UE Context Management service modelled on a real gNB-CU-CP component. It looked at four dimensions: tail-latency predictability, compile-time defect prevention, safety-model overhead, and incremental-adoption cost. Both implementations were built against one frozen specification and driven along identical code paths. That is what lets a language effect be told apart from a difference of design or of skill.

On safety the difference is one of kind and it has a clear boundary. The edge stops where the catalogue says it does. Safe Rust rejects all seven injected memory-safety and data-race classes at compile time. Hazards that C++ can only hope to catch with instrumented tests become programs that cannot be written. All seven boundary classes compile and misbehave in safe Rust as they do in C++. On two of them the C++ tooling reports what Rust's does not. Stated precisely, Rust eliminates data races and aliasing errors. It does not eliminate race conditions, lock-ordering errors, leaks, wrapping arithmetic or the blast radius of a failed worker, and it eliminates nothing inside an `unsafe` block.

On performance the evidence first divides by what is being measured. An end-to-end operation here is 0.250% language. The rest is an operating-system handoff. An end-to-end null bounds nothing about either compiler. The same operation measured apart is 1.6× faster in Rust. At the representative load the paired test resolves 1.3 μs in Rust's favour, too small for a deployment to care about at the median. The far tail on the reported host orders the languages in the same direction. A virtualised host could not settle that percentile. The Rust arms then hold their offered rate to 448 kops/s against 392, and once all four have collapsed C++ sustains 1.7× the throughput of Rust, which spends 1.8× the CPU on each operation it completes. On memory the Rust build carries a few MiB more fixed footprint under both allocators, in the same direction and order of magnitude on all three machines measured. The allocator choice costs about as much on both bare-metal hosts, and an order of magnitude more on the virtualised one this study used first.

What follows is a narrow claim, narrower than this thesis expected to make. An unmodified Rust core can be driven from C++ across a thin seam. The cost on a

real operation is bounded at 10.91%. All `unsafe` is confined to the seam. The core is verified free of it. For stateful, concurrent, partitionable control-plane services of this kind, at the loads where the service still meets its offered rate, the choice is which classes of defect an organisation wants to make unwriteable, and whether it will pay the engineering and learning cost of getting them. Past that point it is a performance decision as well. What the thesis adds is an instrument, not a verdict: a frozen specification, a parity-tested pair of implementations, a measurement framework that reaches past the saturation knee, a decomposition that says how much of what it measures the language is answerable for, and a defect catalogue that maps where the guarantee stops.

Future work answers the limitations one at a time. The immediate priority needs no new machine: vary the transparent-hugepage policy on one host with everything else held fixed. All three hosts agree that the Rust build carries the larger footprint. They disagree about how that compares against the cost of the allocator, which the virtualised host put an order of magnitude higher and neither bare-metal host does. The hugepage policy is the candidate this study can name but not test, because it differed between the hosts and only the system-allocator arm inherits it. Beyond that, replaying anonymised production traces would validate the operation mix. A subsystem with genuinely shared mutable state would exercise the two languages differently than a shared-nothing design does. Evaluating Rust async runtimes against the thread-per-shard model would separate the runtime from the language in the tail result [32]. Measuring what in-process isolation of the two heaps costs would price what protecting an `unsafe`-free core is worth [49]. A longer case study tracking defect density and review effort during an incremental adoption is what would settle the human cost this study can only name.

Statement on the Use of AI Tools

Tool. Anthropic's Claude Opus 5, through the Cursor development environment, is the only generative AI system used in preparing this thesis, from June 2026 until submission. It is named neither as an author, which University of Helsinki policy forbids, nor as a source, which is the author's own choice.

What the assistance covered. It was extensive. The model drafted and revised the running prose and proposed the order in which the arguments are made. It wrote and refactored both implementations of the UE Context Management service, the load generator, the fault-injection catalogue, the interoperability prototype, the analysis code behind the bootstrap intervals and the paired permutation tests, and the figure-generating, build and LaTeX sources. It also suggested candidate literature. The abstract is the one part it must not touch, because it is this programme's maturity test, and University of Helsinki policy never permits a language model in one. The abstract was accordingly written by the author unaided.

What the model did not settle. The research problem, the four research questions, the three hypotheses, and the control on which the study's value rests are the author's. That control is one frozen language-neutral specification and one concurrency model, so language remains the only systematic difference between the two implementations. The author also chose to carry defect classes from outside Rust's compile-time guarantee, which is what keeps RQ2 falsifiable. Every measurement was run by the author, on the machines each result names, and the quantities the thesis quotes reach the page as macros regenerated from the raw logs. The readings are the author's too, including the restraints the evidence imposes: a sweep dominated by cross-thread handoff cannot bound an effect of the language, an unresolved comparison is not a demonstration of equivalence, and an operating point noticed after the data was in hand is described rather than tested.

How the output was checked. Model output was treated as a draft to be falsified, and all of it, prose and code alike, was read by the author before it entered the thesis. Every bibliography entry was resolved against Crossref, DataCite, or the issuing body. Most were then graded by how far the claim made of it could be traced: to the page or clause itself, to a publisher-deposited record carrying it, or no further than confirmed

metadata behind a paywall. The audit records that grade per entry, so an entry whose supporting passage was never opened is listed rather than absorbed. It found four passages that overstated or misplaced their source, one a quotation absent from the work it was attributed to, and all four were corrected. The audit is kept beside the artefacts. Code was checked by running it against a conformance suite whose transcripts are diffed line by line and a fault catalogue that fixes each expected outcome before the defect is injected.

What the author took from it. The model is dependable at mechanical work, at transposing an idiom between two languages or keeping a build honest. It is undependable in one predictable way: it will supply a plausible citation for a claim and a confident sentence where the data supports a hedge. Its fluency is not evidence. The effort saved in drafting is spent again in checking, and the defensible parts of this thesis are the parts checked against a source, a specification, or a measurement.

Responsibility. The author has reviewed the whole of this thesis and is responsible for all of it. Where it is wrong, the error is the author's.

Bibliography

[1] 3GPP. *NG-RAN; Architecture description*. Technical Specification (TS) 38.401. V17.5.0, Release 17. 3rd Generation Partnership Project (3GPP), June 2023. URL: https://www.3gpp.org/ftp/Specs/archive/38_series/38.401/38401-h50.zip.

[2] 3GPP. *NR; NR and NG-RAN Overall description; Stage 2*. Technical Specification (TS) 38.300. V17.5.0, Release 17. 3rd Generation Partnership Project (3GPP), June 2023. URL: https://www.3gpp.org/ftp/Specs/archive/38_series/38.300/38300-h50.zip.

[3] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia. "Understanding O-RAN: Architecture, Interfaces, Algorithms, Security, and Research Challenges". In: *IEEE Communications Surveys & Tutorials* 25.2 (2023), pp. 1376–1411. DOI: 10.1109/COMST.2023.3239220. URL: https://doi.org/10.1109/COMST.2023.3239220.

[4] 3GPP. *NR; Radio Resource Control (RRC); Protocol specification*. Technical Specification (TS) 38.331. V17.5.0, Release 17. 3rd Generation Partnership Project (3GPP), July 2023. URL: https://www.3gpp.org/ftp/Specs/archive/38_series/38.331/38331-h50.zip.

[5] 3GPP. *Study on scenarios and requirements for next generation access technologies*. Technical Report (TR) 38.913. V18.0.0, Release 18. Clause 7.4 sets the control plane latency target, for the transition from a battery-efficient state to the start of continuous data transfer, at 10 ms. 3rd Generation Partnership Project (3GPP), Apr. 2024. URL: https://www.3gpp.org/ftp/Specs/archive/38_series/38.913/38913-i00.zip.

[6] J. Dean and L. A. Barroso. "The Tail at Scale". In: *Communications of the ACM* 56.2 (2013), pp. 74–80. DOI: 10.1145/2408776.2408794. URL: https://doi.org/10.1145/2408776.2408794.

[7] L. Barroso, M. Marty, D. Patterson, and P. Ranganathan. "Attack of the Killer Microseconds". In: *Communications of the ACM* 60.4 (2017), pp. 48–54. DOI: 10.1145/3015146. URL: https://doi.org/10.1145/3015146.

[8] M. Miller. *Trends, challenge, and shifts in software vulnerability mitigation*. BlueHat IL, Microsoft Security Response Center (MSRC). Approximately 70% of Microsoft CVEs are memory-safety issues. 2019. URL: https://github.com/microsoft/MSRC-Security-Research/tree/master/presentations/2019_02_BlueHatIL.

[9] J. Vander Stoep. *Memory Safe Languages in Android 13*. Google Security Blog. Dec. 2022. URL: https://security.googleblog.com/2022/12/memory-safe-languages-in-android-13.html.

[10] S. Klabnik and C. Nichols. *The Rust Programming Language*. 2nd. San Francisco, CA: No Starch Press, 2023. ISBN: 978-1-7185-0310-6. URL: https://openlibrary.org/isbn/9781718503106.

[11] R. Jung, J.-H. Jourdan, R. Krebbers, and D. Dreyer. "RustBelt: Securing the Foundations of the Rust Programming Language". In: *Proceedings of the ACM on Programming Languages* 2.POPL (2018), 66:1–66:34. DOI: 10.1145/3158154. URL: https://doi.org/10.1145/3158154.

[12] A. Balasubramanian, M. S. Baranowski, A. Burtsev, A. Panda, Z. Rakamarić, and L. Ryzhyk. "System Programming in Rust: Beyond Safety". In: *Proceedings of the 16th Workshop on Hot Topics in Operating Systems (HotOS)*. 2017, pp. 156–161. DOI: 10.1145/3102980.3103006. URL: https://doi.org/10.1145/3102980.3103006.

[13] M. Polese, N. Mohamadi, S. D'Oro, L. Bonati, and T. Melodia. "Beyond Connectivity: An Open Architecture for AI-RAN Convergence in 6G". In: *IEEE Communications Magazine* 64.7 (2026), pp. 90–95. DOI: 10.1109/MCOM.001.2500438. URL: https://doi.org/10.1109/MCOM.001.2500438.

[14] ITU-R. *Framework and overall objectives of the future development of IMT for 2030 and beyond*. Recommendation ITU-R M.2160-0. Approved 13 November 2023. Defines the IMT-2030 ("6G") framework and its development timeline. International Telecommunication Union, Radiocommunication Sector, Nov. 2023. URL: https://www.itu.int/rec/R-REC-M.2160/en.

[15] AI-RAN Alliance. *AI-RAN Platform and Infrastructure Orchestrator*. AI-RAN Alliance, Working Group 2 (AI-and-RAN). Version 10-rc1, clause 2.2. Defines AI-and-RAN as one common IT infrastructure accommodating both AI and RAN workloads, which the infrastructure is required to run concurrently, and notes

that the two kinds of workload may have different service-level agreements. Contributors from Ericsson, Nokia, Nvidia, Red Hat, Samsung, SoftBank, zTouch Networks and Northeastern University. Accessed August 2026. Feb. 2026. URL: https://ai-ran.org/documents/AI-RAN-Platform-Infrastructure-Orchestrator.pdf.

[16] J. Vander Stoep. *Rust in Android: Move Fast and Fix Things*. Google Security Blog. Nov. 2025. URL: https://blog.google/security/rust-in-android-move-fast-fix-things/.

[17] A. Rebert and C. Kern. *Secure by Design: Google's Perspective on Memory Safety*. Google Security Engineering, Technical Report. 2024. URL: https://research.google/pubs/secure-by-design-googles-perspective-on-memory-safety/.

[18] R. N. M. Watson, J. Baldwin, D. Chisnall, T. Chen, J. Clarke, B. Davis, N. W. Filardo, B. Gutstein, G. Jenkinson, B. Laurie, A. Mazzinghi, S. W. Moore, P. G. Neumann, H. Okhravi, A. Richardson, A. Rebert, P. Sewell, L. Tratt, M. Vijayaraghavan, H. Vincent, and K. Witaszczyk. "It Is Time to Standardize Principles and Practices for Software Memory Safety". In: *Communications of the ACM* 68.2 (2025), pp. 40–45. DOI: 10.1145/3708553. URL: https://doi.org/10.1145/3708553.

[19] Cybersecurity and Infrastructure Security Agency. *The Case for Memory Safe Roadmaps: Why Both C-Suite Executives and Technical Experts Need to Take Memory Safe Coding Seriously*. Cybersecurity Information Sheet. CISA, NSA, FBI, ASD's ACSC, CCCS, NCSC-UK, NCSC-NZ, CERT NZ, Dec. 2023. URL: https://www.cisa.gov/resources-tools/resources/case-memory-safe-roadmaps.

[20] Office of the National Cyber Director. *Back to the Building Blocks: A Path Toward Secure and Measurable Software*. Technical Report. Washington, DC: The White House, Office of the National Cyber Director (ONCD), Feb. 2024. URL: https://bidenwhitehouse.archives.gov/wp-content/uploads/2024/02/Final-ONCD-Technical-Report.pdf.

[21] European Parliament and Council of the European Union. *Regulation (EU) 2024/2847 on horizontal cybersecurity requirements for products with digital elements (Cyber Resilience Act)*. Official Journal of the European Union, OJ L,

2024/2847. Nov. 2024. URL: https://eur-lex.europa.eu/eli/reg/2024/2847/oj.

[22] Dell'Oro Group. *RAN Mostly Stable in 3Q 2025*. Dell'Oro Group, RAN Quarterly Report summary. The five largest RAN suppliers by worldwide revenue (Huawei, Ericsson, Nokia, ZTE, Samsung) accounted for 96% of the 1Q25–3Q25 RAN market, up from 95% in 2024. Accessed August 2026. Nov. 2025. URL: https://www.delloro.com/news/ran-mostly-stable-in-3q-2025/.

[23] Dell'Oro Group. *RAN Market Stabilized in 2025*. Dell'Oro Group, RAN Quarterly Report summary. Full-year 2025 preliminary findings: supplier rankings unchanged, and overall market concentration measured by the Herfindahl–Hirschman Index reached a ten-year high. Accessed August 2026. Feb. 2026. URL: https://www.delloro.com/news/ran-market-stabilized-in-2025/.

[24] J. Almodovar and A. Sultan. *The SA1 road to 6G*. 3GPP Highlights, Issue 09. Written by the chair and the secretary of 3GPP WG SA1. Release 20 carries the 6G studies and Release 21 the normative work. ITU-R sets early 2029 for technology proposals and mid-2030 for the submitted specifications. Accessed August 2026. Dec. 2024. URL: https://www.3gpp.org/news-events/3gpp-news/sa1-6g-road.

[25] B. Qin, Y. Chen, Z. Yu, L. Song, and Y. Zhang. "Understanding Memory and Thread Safety Practices and Issues in Real-World Rust Programs". In: *Proceedings of the 41st ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*. 2020, pp. 763–779. DOI: 10.1145/3385412.3386036. URL: https://doi.org/10.1145/3385412.3386036.

[26] A. Meneely, A. Green, T. Jaafari, M. Fluet, and B. Keller. ""Just Use Rust": A Best-Case Historical Study of Open Source Vulnerabilities in C". In: *Proceedings of the 3rd IEEE/ACM International Workshop on Software Vulnerability Management (SVM)*. 2025, pp. 25–32. DOI: 10.1109/SVM66695.2025.00008. URL: https://doi.org/10.1109/SVM66695.2025.00008.

[27] 3GPP. *Service requirements for the 5G system; Stage 1*. Technical Specification (TS) 22.261. V18.6.0, Release 18. 3rd Generation Partnership Project (3GPP), Mar. 2022. URL: https://www.3gpp.org/ftp/Specs/archive/22_series/22.261/22261-i60.zip.

[28] 3GPP. *NG-RAN; E1 Application Protocol (E1AP)*. Technical Specification (TS) 38.463. V17.0.0, Release 17. 3rd Generation Partnership Project (3GPP), Apr. 2022. URL: https://www.3gpp.org/ftp/Specs/archive/38_series/38.463/38463-h00.zip.

[29] R. Karlbäck and A. Orö. "Holistic View on Alternative Programming Languages for Radio Access Network Applications in Cloud and Embedded Deployments". Carried out at Ericsson AB. Report LIU-IDA/LITH-EX-A–2021/023–SE. Master's thesis. Linköping University, Department of Computer and Information Science, 2021. URL: https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-176307.

[30] V. Olsson. "Rust Programming Language as an Alternative to C for RAN Management Applications". Carried out at Ericsson AB. Master's thesis. KTH Royal Institute of Technology, School of Electrical Engineering and Computer Science, 2023. URL: https://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-325822.

[31] J. Olsson and I. Stenström. "Rust Async for Thread-per-Core Packet Processing". Carried out at Ericsson AB. Report LIU-IDA/LITH-EX-A–23/052–SE. Master's thesis. Linköping University, Department of Computer and Information Science, 2023. URL: https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-197029.

[32] A. Arvidsson and M. Kaller. "Evaluating Asynchronous Programming Frameworks for 5G Control Plane Procedures". Carried out at Ericsson AB. Report LIU-IDA/LITH-EX-A–25/055–SE. Master's thesis. Linköping University, Department of Computer and Information Science, 2025. URL: https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-214746.

[33] C. Emberg and A. Wadsten. "Attainable Safety and Long-Term Resource-Efficiency Using Rust: Evaluating the Viability of Rust in Terms of Development Effort and Performance for the Decompilation of ASN.1 Messages". Carried out at Ericsson, Linköping. Report LIU-IDA/LITH-EX-A–23/078–SE. Master's thesis. Linköping University, Department of Computer and Information Science, 2023. URL: https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-195552.

[34] R. Gjomemo, P. H. Phung, E. Ballou, K. S. Namjoshi, V. N. Venkatakrishnan, and L. Zuck. "Leveraging Static Analysis Tools for Improving Usability of Memory Error Sanitization Compilers". In: *Proceedings of the 2016 IEEE International Conference on Software Quality, Reliability and Security (QRS)*. Two of

the six authors are affiliated with Nokia. 2016, pp. 323–334. DOI: 10.1109/QRS.2016.44. URL: https://doi.org/10.1109/QRS.2016.44.

[35] P. Enberg, A. Rao, and S. Tarkoma. "The Impact of Thread-per-Core Architecture on Application Tail Latency". In: *Proceedings of the 2019 ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS)*. 2019, pp. 1–8. DOI: 10.1109/ANCS.2019.8901874. URL: https://doi.org/10.1109/ANCS.2019.8901874.

[36] G. Tene. *HdrHistogram: A High Dynamic Range Histogram*. Project documentation and reference implementation. Section "Corrected vs. raw value recording calls" documents the `recordValueWithExpectedInterval` correction applied when a response time exceeds the expected interval between requests. Accessed August 2026. URL: https://github.com/HdrHistogram/HdrHistogram.

[37] R. Natella, D. Cotroneo, and H. S. Madeira. "Assessing Dependability with Software Fault Injection: A Survey". In: *ACM Computing Surveys* 48.3 (2016), 44:1–44:55. DOI: 10.1145/2841425. URL: https://doi.org/10.1145/2841425.

[38] Z. Li, J. Wang, M. Sun, and J. C. S. Lui. "Detecting Cross-Language Memory Management Issues in Rust". In: *Computer Security – ESORICS 2022*. 2022, pp. 680–700. DOI: 10.1007/978-3-031-17143-7_33. URL: https://doi.org/10.1007/978-3-031-17143-7_33.

[39] A. Georges, D. Buytaert, and L. Eeckhout. "Statistically Rigorous Java Performance Evaluation". In: *Proceedings of the 22nd ACM SIGPLAN Conference on Object-Oriented Programming Systems, Languages and Applications (OOPSLA)*. 2007, pp. 57–76. DOI: 10.1145/1297027.1297033. URL: https://doi.org/10.1145/1297027.1297033.

[40] B. Efron and R. J. Tibshirani. *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993. ISBN: 978-0-412-04231-7.

[41] S. Holm. "A Simple Sequentially Rejective Multiple Test Procedure". In: *Scandinavian Journal of Statistics* 6 (1979), pp. 65–70. URL: https://www.jstor.org/stable/4615733.

[42] T. Yin. *Lizard: An extensible cyclomatic complexity analyzer for many programming languages*. Version 1.23.0. Accessed August 2026. 2026. URL: https://github.com/terryyin/lizard.

[43] X. Zhou, V. Leis, X. Yu, and M. Stonebraker. "OLTP Through the Looking Glass 16 Years Later: Communication Is the New Bottleneck". In: *15th Annual Conference on Innovative Data Systems Research (CIDR 2025)*. 2025. URL: https://www.vldb.org/cidrdb/papers/2025/p17-zhou.pdf.

[44] J. Bornholt, R. Joshi, V. Astrauskas, B. Cully, B. Kragl, S. Markle, K. Sauri, D. Schleit, G. Slatton, S. Tasiran, J. Van Geffen, and A. Warfield. "Using Lightweight Formal Methods to Validate a Key-Value Storage Node in Amazon S3". In: *Proceedings of the ACM SIGOPS 28th Symposium on Operating Systems Principles (SOSP)*. 2021, pp. 836–850. DOI: 10.1145/3477132.3483540. URL: https://doi.org/10.1145/3477132.3483540.

[45] Y. Zhang, K. Zhang, G. Liu, Y. Cai, and S. Qin. "Two Birds One Stone: Effective Static Detection of Resource and Communication Deadlocks in Rust Programs". In: *Automated Software Engineering* 33.2, 65 (2026). Extends the preprint arXiv:2401.01114. DOI: 10.1007/s10515-026-00614-z. URL: https://doi.org/10.1007/s10515-026-00614-z.

[46] A. Sundvisson and S. Hofer. "User-Space Device Drivers on Embedded Linux: A Comparative Case Study of C and Rust". Bachelor's thesis. Blekinge Institute of Technology, Department of Software Engineering, 2026. URL: https://urn.kb.se/resolve?urn=urn:nbn:se:bth-29775.

[47] K. R. Fulton, A. Chan, D. Votipka, M. Hicks, and M. L. Mazurek. "Benefits and Drawbacks of Adopting a Secure Programming Language: Rust as a Case Study". In: *Proceedings of the 17th Symposium on Usable Privacy and Security (SOUPS)*. USENIX Association, 2021, pp. 597–616. URL: https://www.usenix.org/conference/soups2021/presentation/fulton.

[48] W. Crichton and S. Krishnamurthi. "Profiling Programming Language Learning". In: *Proceedings of the ACM on Programming Languages* 8.OOPSLA1 (2024), pp. 29–54. DOI: 10.1145/3649812. URL: https://doi.org/10.1145/3649812.

[49] E. Rivera, S. Mergendahl, H. Shrobe, H. Okhravi, and N. Burow. "Keeping Safe Rust Safe with Galeed". In: *Proceedings of the 37th Annual Computer Security Applications Conference (ACSAC)*. 2021, pp. 824–836. DOI: 10.1145/3485832.3485903. URL: https://doi.org/10.1145/3485832.3485903.

[50] P. Kirth, M. Dickerson, S. Crane, P. Larsen, A. Dabrowski, D. Gens, Y. Na, S. Volckaert, and M. Franz. "PKRU-Safe: Automatically Locking Down the Heap Between Safe and Unsafe Languages". In: *Proceedings of the 17th European Conference on Computer Systems (EuroSys)*. 2022, pp. 132–148. DOI: 10.1145/3492321.3519582. URL: https://doi.org/10.1145/3492321.3519582.

[51] A. N. Evans, B. Campbell, and M. L. Soffa. "Is Rust Used Safely by Software Developers?" In: *Proceedings of the 42nd International Conference on Software Engineering (ICSE)*. 2020, pp. 246–257. DOI: 10.1145/3377811.3380413. URL: https://doi.org/10.1145/3377811.3380413.